

MTH 446: Lecture # 10

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Lecture Span

1. Limits of complex functions

Recap

You may have to extend functions to the extended complex plane. An example

$$\tilde{f}(z) = \begin{cases} 1/z & z \neq 0 \vee \infty \\ 0 & z = \infty \\ \infty & z = 0 \end{cases} \quad (1)$$

Use algebraic properties of limits. For the third case, prove that $1/f(z) = z \rightarrow 0$. Clearly, this is true.

Also, we need to show that $z \rightarrow z_0$, where $z_0 \in G = \text{dom}(f)$. In other words, z_0 needs to be in the domain of f . For the second case, $|z| > 1/\delta \implies |1/z - 0| = 1/|z| < \epsilon \iff |z| > 1/\epsilon$. Therefore, let $\delta = \epsilon$, then we see that

$$\lim_{z \rightarrow 0} (1/z) = \infty \quad (2)$$

Therefore, $\tilde{f} \in C(\overline{\mathbb{C}})$ (cont. in the extended complex plane).

Example

(Möbius transformation) Defined by

$$M(z) = \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{C} \quad (3)$$

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq 0 \quad (4)$$

Therefore, if $M(z) = 1/z = (0 \cdot z + 1)/(1 \cdot z + 0)$ which is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (5)$$

Extend M to the extended complex plane. Consider

$$M_1 \circ M_2 = M_1(M_2(z)) \quad (6)$$

With $\det M_1 \neq 0$ and $\det M_2 \neq 0$. Let $c \neq 0$, then we can extend M to the extended complex plane:

$$\tilde{M}(z) = \begin{cases} \frac{az+b}{cz+d} & z \neq \infty, z \neq -d/c \\ \infty & z = -d/c \\ a/z & z = \infty \end{cases} \quad (7)$$

Case 1: if $z \neq \infty$ and $z \neq -d/c$. Then $az + b$ and $cz + d$ are in $C(\overline{\mathbb{C}})$. Moreover, $cz + d \neq 0$. Then, by the algebraic properties of cont. functions, then $\tilde{M}(z)$ is cont.

Case 2: if $z = -d/c$. Firstly, $-d/c \in \overline{\mathbb{C}}$. Next, write

$$\lim_{z \rightarrow -d/c} \tilde{M}(z) = \infty \iff \lim_{z \rightarrow -d/c} 1/\tilde{M}(z) = 0 \quad (8)$$

Then

$$\lim 1/\tilde{M} = \lim \frac{cz + d}{az + b} \quad (9)$$

$$\frac{cz + d}{az + b} \rightarrow \frac{0}{-ad/c + b} = 0 \iff ad - bc \neq 0 \quad (10)$$

But, this is the definition of the Mobius transformation, therefore, $1/\tilde{M}(z) \rightarrow 0 \iff \tilde{M}(z) \rightarrow \infty$.

Case 3: $z_0 \rightarrow \infty$. Then, evaluate

$$\lim \frac{az + b}{cz + d} = \lim \frac{a + b/z}{c + d/z} \rightarrow a/c \quad (11)$$

Here, we used the algebraic properties of cont. functions.

Next, if $c = 0$, then $ad - bc = ad \neq 0$ which means that $a \neq 0$ and $d \neq 0$. Therefore,

$$M(z) = a/d \cdot z + b/d \quad (12)$$

Then

$$M(z) = Az + B, A \neq 0 \quad (13)$$

We define this transformation onto the extended complex plane. in other words,

$$\tilde{M}(z) = \begin{cases} Az + B & z \neq \infty \\ \infty & z = \infty \end{cases} \quad (14)$$

Case 1: obvious.

Case 2:

$$\lim 1/M(z) = \lim \frac{1}{Az + B} = \lim \frac{1/z}{A + B/z} \rightarrow 0/A = 0 \quad (15)$$

Therefore, $\lim \tilde{M}(z) = \infty$ and $\tilde{M}(\infty) = \infty$. Therefore, $\tilde{M} \in \overline{\mathbb{C}}$.

Midterm:

1. 5 problems 20 points each, no cheat sheet.
2. problem 1: general information about complex numbers. Conjugate, modulus. $|z|^2 = z\bar{z}$.
3. problem 2: roots of complex number. Principal root
4. problem 3: Problem of limits
5. problem 4: Problem related to Demoire formula
6. problem 5: problem related to open set, closed set, interior, boundary (D intersects with outside and inside for all ϵ). Find closure. Accumulation points.

Complex Derivative

Definition 0.1. f is complex differentiable at $z = z_0$ if the following exists:

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \quad (16)$$

$f'(z_0)$ is the complex derivative at $z = z_0$. Of course, D is open in $\mathbb{C}$ and $f : D \rightarrow \mathbb{C}$.

Of course,

1. $(\alpha f + \beta g)' = \alpha f' + \beta g'$
2. $(fg)' = f'g + fg'$
3. Quotient rule is the same, if $g(z_0) \neq 0$.

Cauchy-Riemann equations

Suppose that $f'(z_0)$ exists. Then, (recall that $f = u + iv$)

$$f'(z_0) = u_x + iv_x = v_y - iu_x, \text{ approaching limit from real and complex axis} \quad (17)$$

Therefore, the Cauchy-Riemann equations.

$$u_x = v_y, \quad u_y = -v_x \quad (18)$$